

# Saaras Pakanati

Cincinnati, Ohio, USA

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## Education

**University of Cincinnati, BS Mechanical Engineering**

Minor in Mathematics

Advisor: Prof. Kishan Bellur

August 2021 – Present

GPA 3.95/4.00

## Research Experience

**Undergraduate Research Assistant**

UC Lab for Interfacial Dynamics, advised by Dr. Kishan Bellur

Dept. of Mechanical & Materials Engineering, University of Cincinnati

August 2024 – Present

- **Numerical & Experimental study of multiphase interfaces and associated phenomenon.**

- Modeled multiscale cryogenic fuel evaporation using custom UDFs in ANSYS Fluent.
- Designed & built an interferometry setup to study instabilities in liquid thin films.
- Assisting in the study of the effect of temperature-sensitive surfactants on interfacial dynamics.

## Peer-Reviewed Publications

\* denotes equal contribution

2. A. Sarchami, **S. Pakanati**, A. Yasin, M. Jog, K. Bellur, *An experimental study of interfacial dynamics control using temperature-sensitive surfactants*, Langmuir, 2026 (Accepted)

1. A. Yasin\*, **S. Pakanati\***, K. Bellur, *A multiscale CFD model of evaporating Hydrogen menisci: Incorporating subgrid thin-film dynamics and in situ accommodation coefficients*, Fuels, 2026 <https://doi.org/10.3390/fuels7010003>

## Conference Presentations

2. A. Sarchami, **S. Pakanati**, T. Enam, A. Yasin, K. Bellur, *Stability and Contact Line Dynamics of Evaporating Thin Liquid Films*, 11<sup>th</sup> ASTFE Thermal and Fluids Engineering Conference, Tempe, Arizona, March 9–12, 2026

1. A. Sarchami, **S. Pakanati**, K. Bellur, *Dynamic Optical Measurement of Evaporating Liquid Films*, ASME International Mechanical Engineering Congress & Exposition, Memphis, Tennessee, November 16–20, 2025 (Poster)

## Honors, Awards, and Fellowships

**Technical Excellence in Senior Design Project – 2<sup>nd</sup> place**, Awarded by the College of Engineering and Applied Science 2026

**Merchant, M. Eugene Scholarship**, Awarded by the College of Engineering and Applied Science 2026

**Armstrong Discovery Fellowship (2×)**, Awarded by the Armstrong Institute for Space, Technology, and Research 2025 & 2026

**CEAS Annual Scholar – Undergraduate Research**, Awarded by the College of Engineering and Applied Science 2025

**Undergraduate Research co-op Fellowship (2×)**, Awarded by the College of Engineering and Applied Science 2024 & 2025

**UC Calculus Contest – 2<sup>nd</sup> place**, Awarded by Cengage Publishing & UC Department of Mathematical Sciences 2022

**CEAS International Outreach Scholarship**, Awarded by the College of Engineering and Applied Science 2021

**University of Cincinnati Global Scholarship**, Awarded by the University of Cincinnati 2021

## Teaching Experience

**Grader**

Dept. of Mechanical & Materials Engineering, University of Cincinnati

August 2025 – December 2025

- Course(s): MECH 2010 (Thermodynamics)
- Managed assignments and proctored exams for 150+ students.

- Led in-class review sessions for Calculus I and II.
- Tutored students in 15 different Engineering, Mathematics, Physics, and Chemistry courses.

## Industry Experience

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**New Product Development Mechanical Engineering co-op**  
MSA Safety, Cranberry Township, Pennsylvania

January 2024 – April 2024

- Executed Finite Element Analysis in Ansys to validate design decisions for safety devices.
- Optimized prototype testing of Technology Readiness Level (TRL) 1 gas detection technologies by developing and applying Design of Experiments (DOE) methods using Minitab.
- Collaborated with Electrical and Software teams for cross-functional design.

**Engineering co-op**  
Richards Industrials, Cincinnati, Ohio

August 2022 – December 2022 & May 2023 – August 2023

- Tested critical load-bearing components using destructive and non-destructive methods.
- Created technical drawings for shop floor fixtures using SolidWorks, collaborating with machinists.
- Developed tools and reports for engineering calculations, process optimization, and quality analysis.

**Software Intern**  
Visual Collaboration Technologies, Inc., Bengaluru, Karnataka, India

June 2022 – July 2022

- Created a visualization tool for a CFD study using the dash framework in Python.
- Collaborated in an Agile development framework to code, following PEP8 standards.

## Skills

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**Simulation:** Ansys (Fluent, Static Structural, LS Dyna), Siemens simulation suite (Siemens NX, StarCCM+), OpenFOAM, ParaView, SolidWorks, LAMMPS

**Programming:** Python, MATLAB, C, C++, Bash, R Language, Minitab, LabView,  $\text{\LaTeX}$

**Experimental:** Optical interferometry, image processing and analysis, flow control, semiconductor manufacturing, Design of Experiments

## Certifications & Licenses

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**OASiS Foundations Program**, *University of Cincinnati* – 2025 | [link to certificate](#)

**MATLAB Fundamentals**, *Mathworks* – 2023 | [link to certificate](#)

**IBM Data Science Practitioner**, *IBM* – 2022 | [link to certificate](#)

**Six Sigma Yellow Belt**, *The Council for Six Sigma Certification* – 2022 | Credential ID: h7L0bACNv5

**NX – CAD Design Associate**, *Siemens* – 2022 | Credential ID: 6763-164-884-2231

## Extra-Curricular

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**President, Undergraduate Research Society**  
University of Cincinnati

September 2021 – May 2026

- President (2025–26), VP (2024–25) and Events Chair (2022–23 & 2023–24)
- Organized meetings with the executive team; managing presentations and logistics for regular outreach.
- Gave multiple presentations on undergraduate research at university classes, general meetings, and academic organizations.

**Volunteer Tutor, Bearcat Buddies**  
Cincinnati, Ohio

October 2021 – December 2023

- Tutored students in the Cincinnati Public School system and assisted in the improvement of the content covered with the local in-charge.